GEOSPATIAL REGRESSION AND MACHINE LEARNING: Part I

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Course outline

Part I: Introduction to geostatistics and spatial linear models

Part II: Random forests for geospatial data

Part III: Neural networks for geospatial data

Part IV: Software demonstration

Course outline

Part I: Introduction to geostatistics and spatial linear models

Part II: Random forests for geospatial data

- Part IV a: Software demonstration of random forests for spatial analysis in R

Part III: Neural networks for geospatial data

- Part IV b: Software demonstration of neural nets for spatial analysis in Python

Overview of Part I

Introduction to geostatistics

Exploratory data analysis
- Maps and variograms

Gaussian Processes (GP) and spatial linear regression
- Estimation and prediction (kriging)
- Spatial linear mixed effect models

Big spatial data
- Computing challenges
- Fast alternatives (Nearest Neighbor Gaussian Process)

What is spatial data

Any data with some geographical information

Common sources of spatial data:

climatology, forestry, ecology, environmental health, disease epidemiology, real estate marketing

Other examples where spatial need not refer to space on earth:

Neuroimaging (data for each voxel in the brain)

Genetics (position along a chromosome)

Spatial transcriptomics (gene expression on slides)

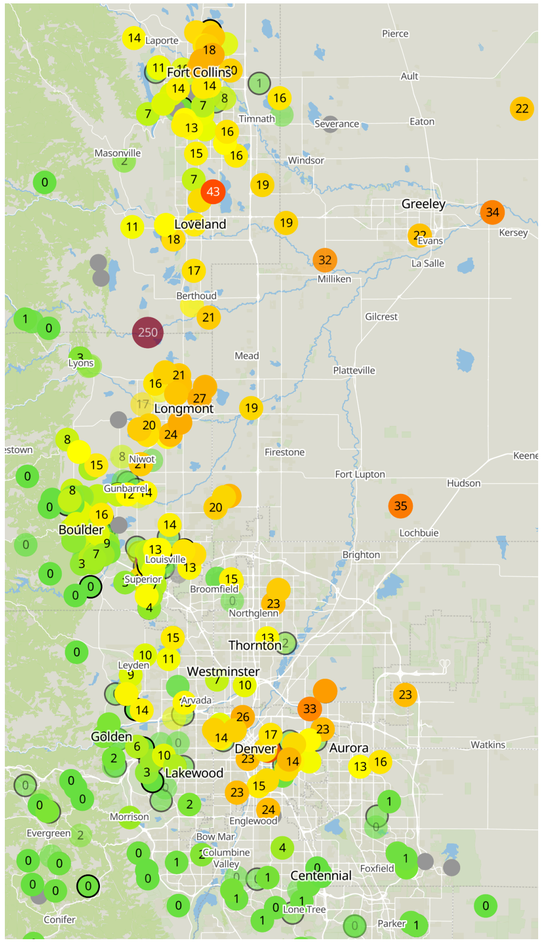
Geostatistics

Each observation (data unit) is associated with a geographical location (latitude-longitude)

Data represents a sample from a continuous spatial domain

Often displayed on a map

Referred to as geocoded/ geostatistical/ point referenced data



$PM_{2.5}$ ($\mu g/m^3$) in Colorado
on Nov 12, 2024 from PurpleAir.com

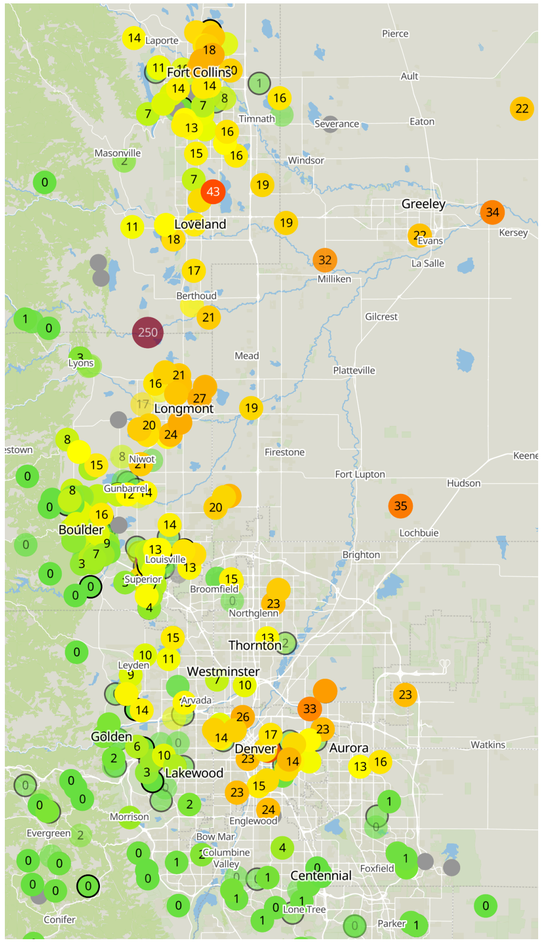
Geostatistics

Point referenced data:

Data collected at locations $s_1, \ldots, s_n$

$Y_i = Y(s_i)$: scalar response at location s_i

$X_i = X(s_i)$: $d \times 1$ vector of covariates (explanatory variables)



$PM_{2.5}$ ($\mu g/m^3$) in Colorado on Nov 12, 2024 from PurpleAir.com

Geostatistics

Point referenced data:

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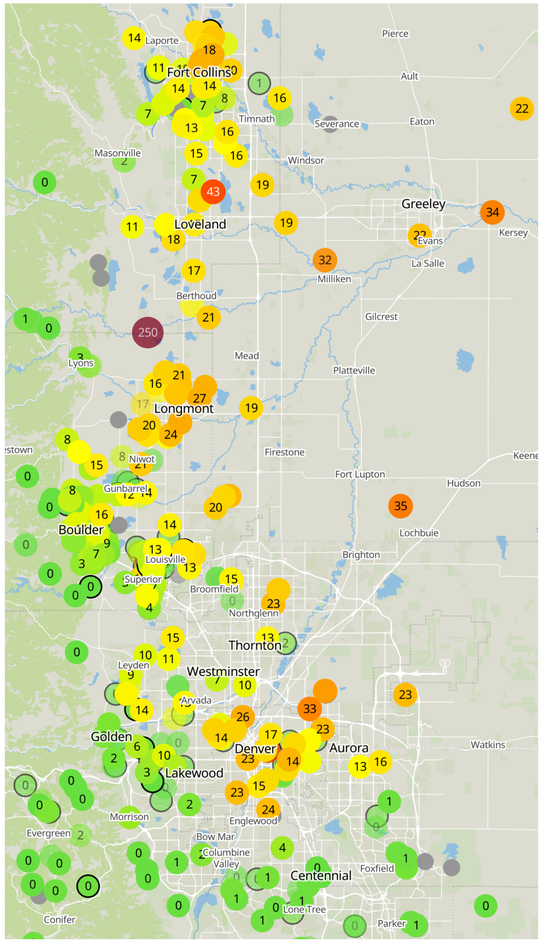
$X_i = X(s_i)$: $d \times 1$ vector of covariates (explanatory variables)

Objectives:

Predict Y at any location without data

Understand spatial patterns in Y

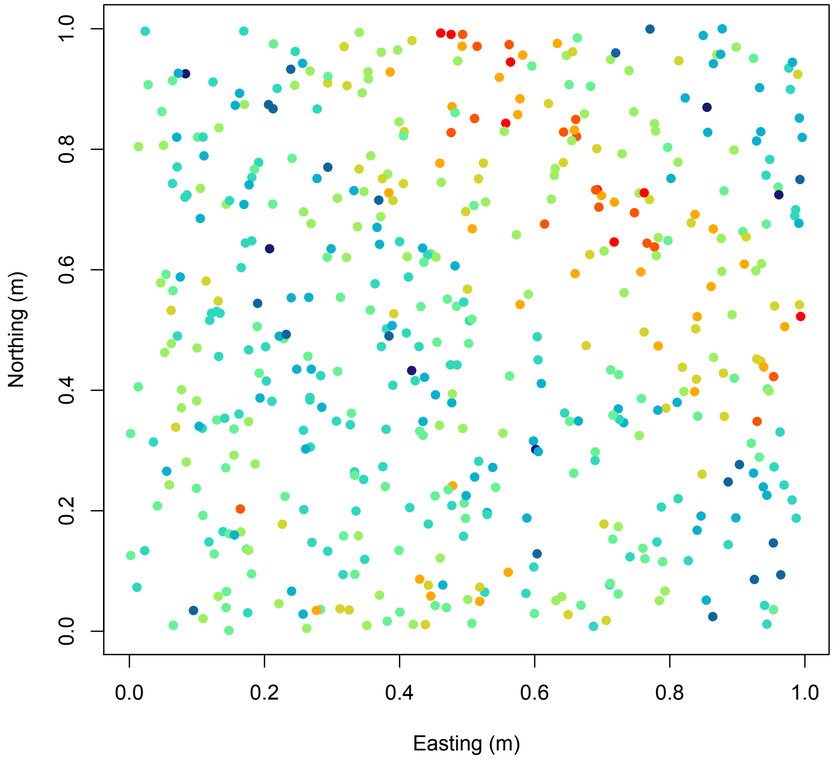
Understand relationship between X and Y



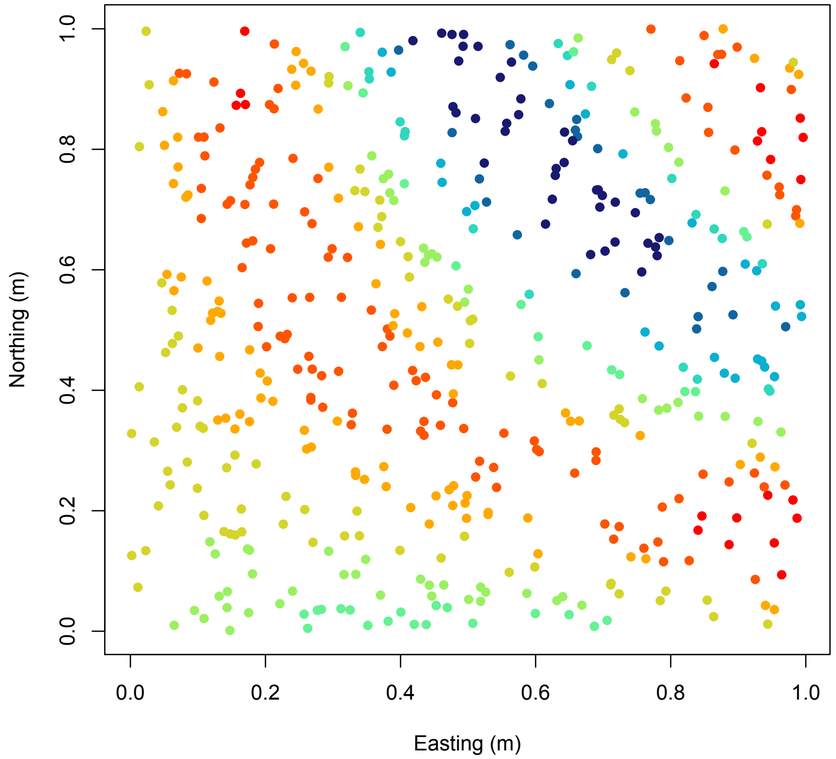
$PM_{2.5}$ ($\mu g/m^3$) in Colorado on Nov 12, 2024 from PurpleAir.com

Exploratory data analysis (EDA): Plotting the data

Point plots help to visualize the exact data where they are observed



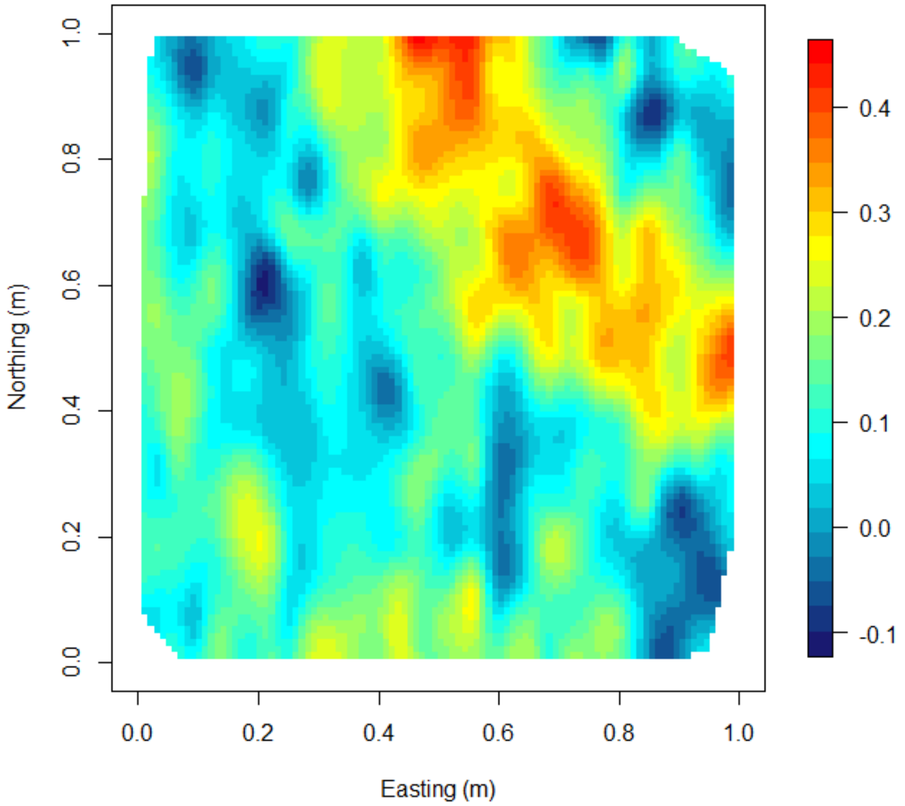
$Y(s_i)$



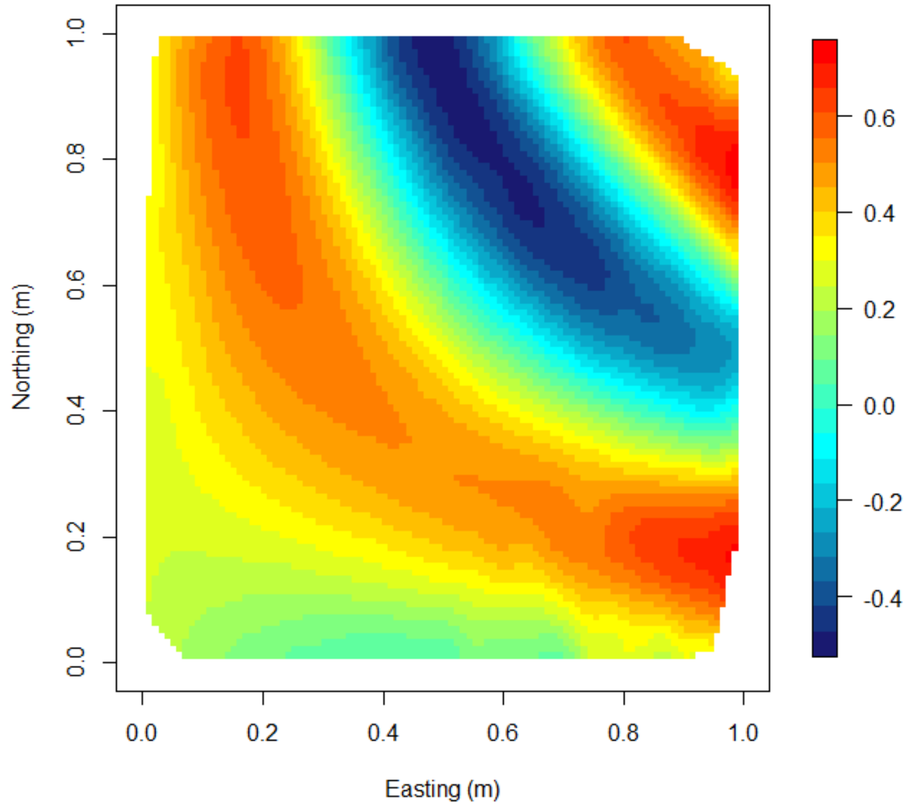
$X(s_i)$

Exploratory data analysis (EDA): Plotting the data

Surface plots (interpolated data) are often better help understand spatial patterns



$Y(s)$



$X(s)$

What's so special about spatial?

Linear regression model: $Y(s_i) = X(s_i)'\beta + \epsilon(s_i)$

$\epsilon(s_i)$ are iid $N(0,\tau^2)$ errors

$Y = (Y(s_1), Y(s_2), \ldots, Y(s_n))'; X = (X(s_1)', X(s_2)', \ldots, X(s_n)')'$

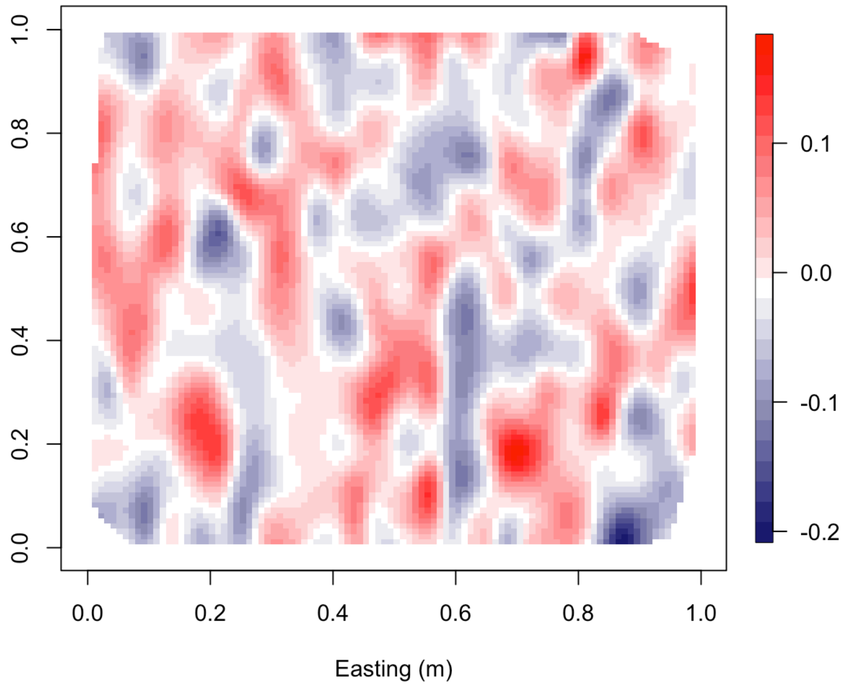
Inference: $\hat{\beta} = (X'X)^{-1}X'Y \sim N(\beta, \tau^2(X'X)^{-1})$

Prediction at new location s_0: $\widehat{Y(s_0)} = X(s_0)'\hat{\beta}$

Although the data is spatial, this is an ordinary linear regression model

Residual plots

Surface plots of residuals $y(s) - \hat{y}(s)$ help identify residual spatial patterns not explained by the covariates



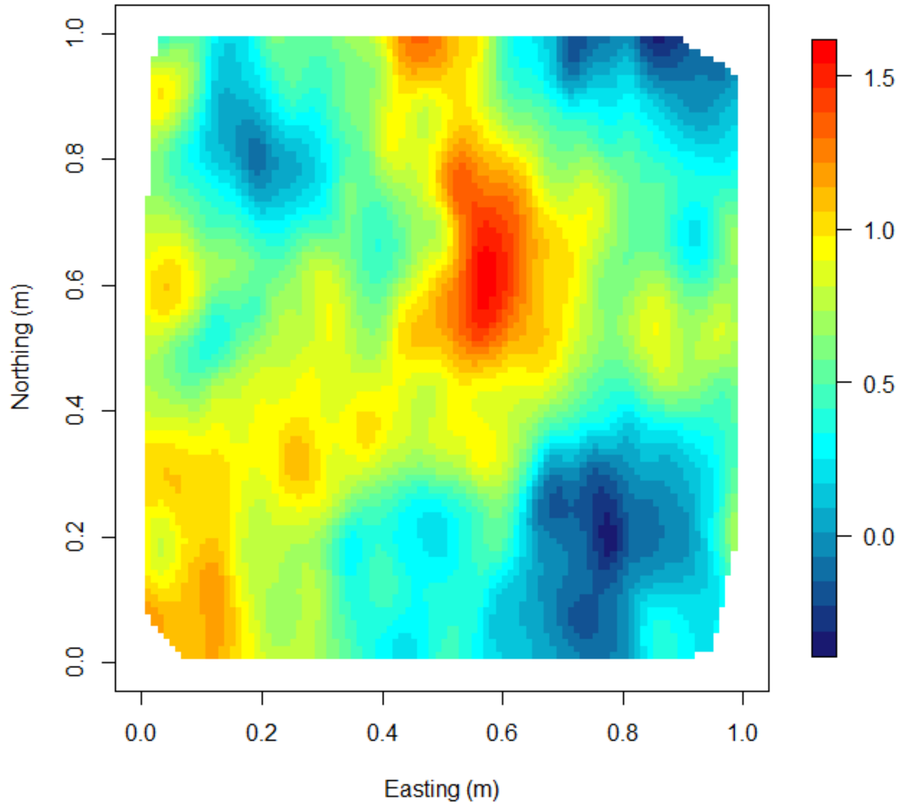
Surface plot of residuals

Surface plot of residuals not showing any large scale spatial patterns

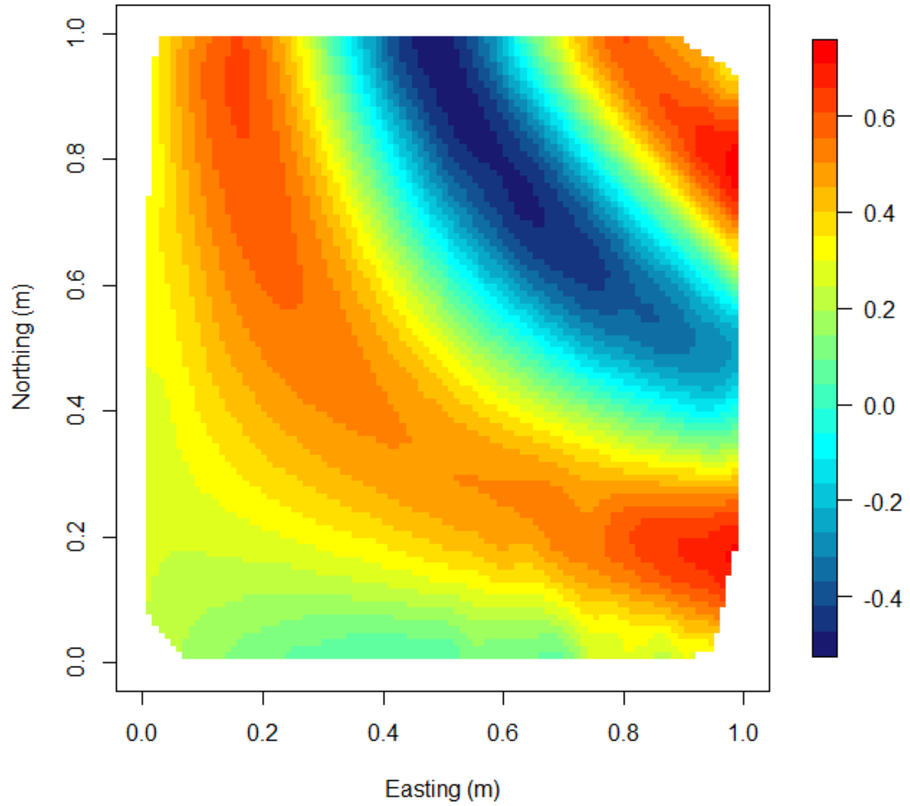
The covariate $X(s)$ seems to explain all spatial variation in the response $Y(s)$

When does such a non-spatial analysis suffice?

Another dataset



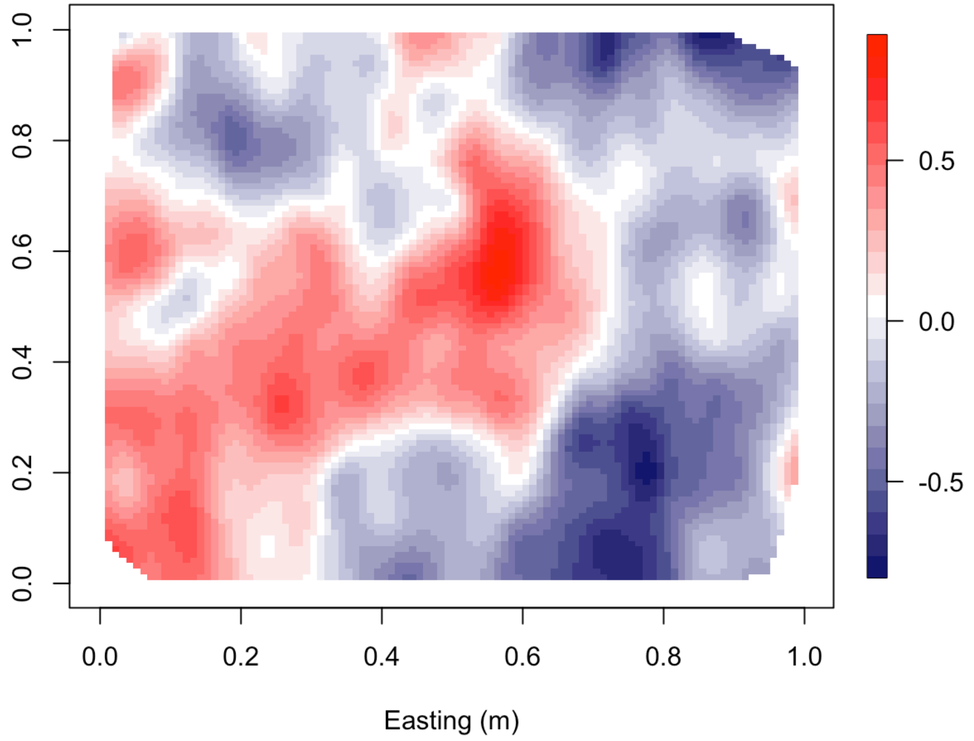
Dataset 2: $Y(s)$



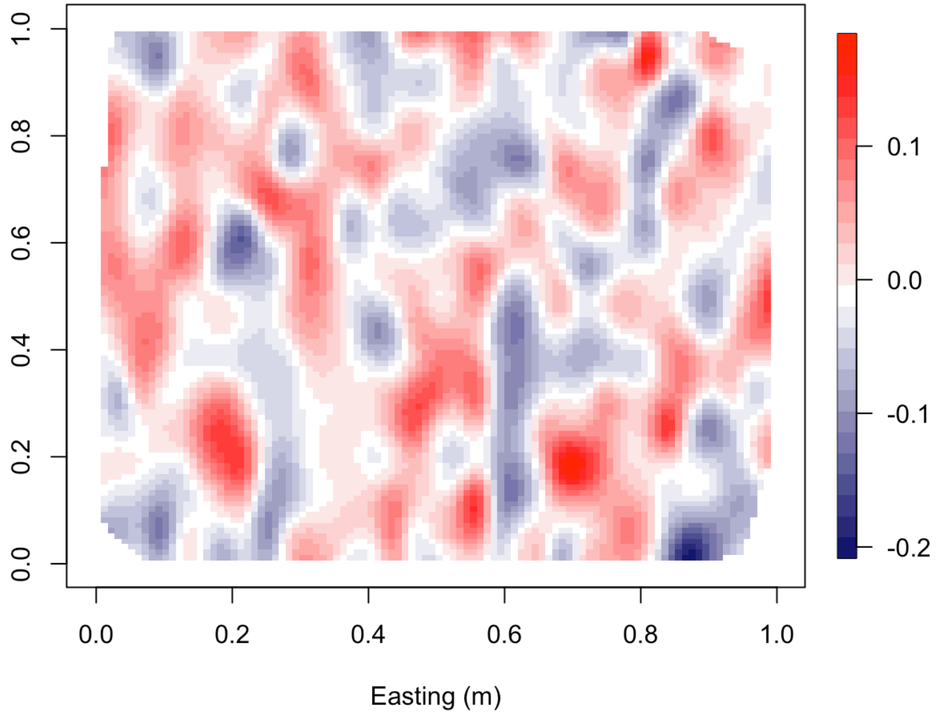
Same $X(s)$

Another dataset

Linear regression: $y(s_i) = \beta_0 + x(s_i)\beta_1 + \epsilon(s_i)$



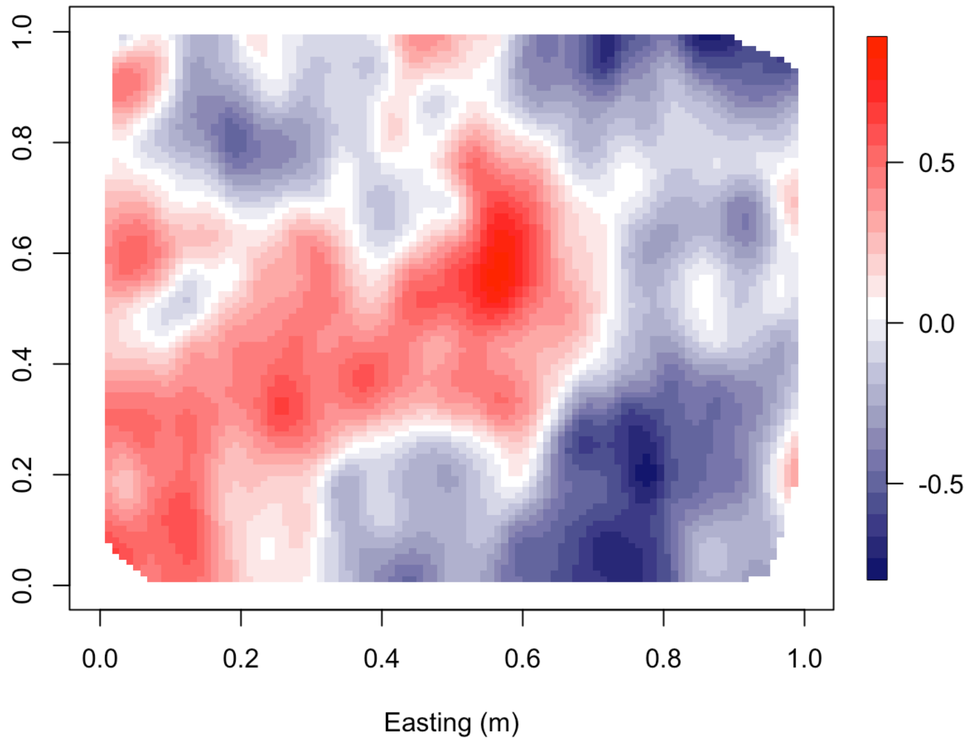
Dataset 2: Residual plot



Dataset 1: Residual plot

Another dataset

Linear regression: $y(s_i) = \beta_0 + x(s_i)\beta_1 + \epsilon(s_i)$



Dataset 2: Residual plot

Strong residual spatial pattern

The covariate $X(s)$ does not explain all spatial variation in the response $Y(s)$

Besides eyeballing residual surfaces, how to do more formal EDA to identify spatial pattern ?

Semi-Variogram

First Law of Geography: *"Everything is related to everything else, but near things are more related than distant things."* – Waldo Tobler

$Y(s_1)$ and $Y(s_2)$ should be more similar if s_1 is near s_2

$(Y(s_1) - Y(s_2))^2$ should be small when $\|s_1 - s_2\|$ is small and increase as $\|s_1 - s_2\|$ increases

Can this be formalized to identify spatial pattern in data?

Variogram and Semi-Variogram

Empirical variogram at distance h: Average of $(Y(s_1) - Y(s_2))^2$ for all pairs s_1, s_2 such that $s_1 - s_2 \approx h$

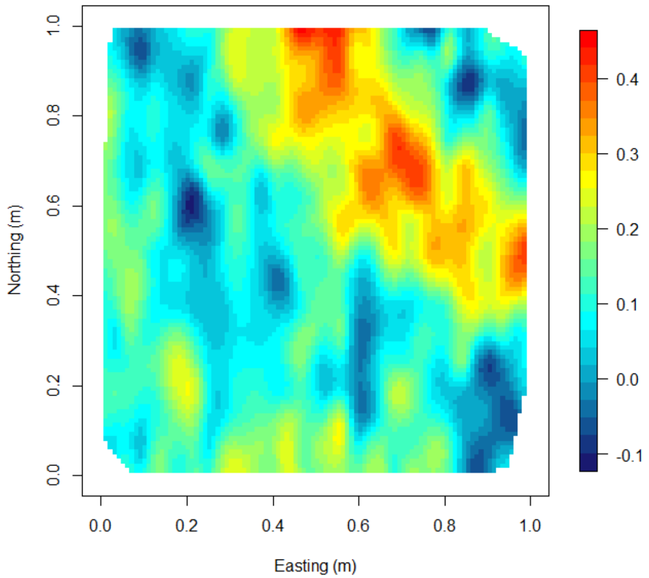
Empirical semi-variogram: $\gamma(h) = \frac{1}{2}$ Empirical variogram at h

For spatial data, the $\gamma(h)$ is expected to roughly increase with the distance h

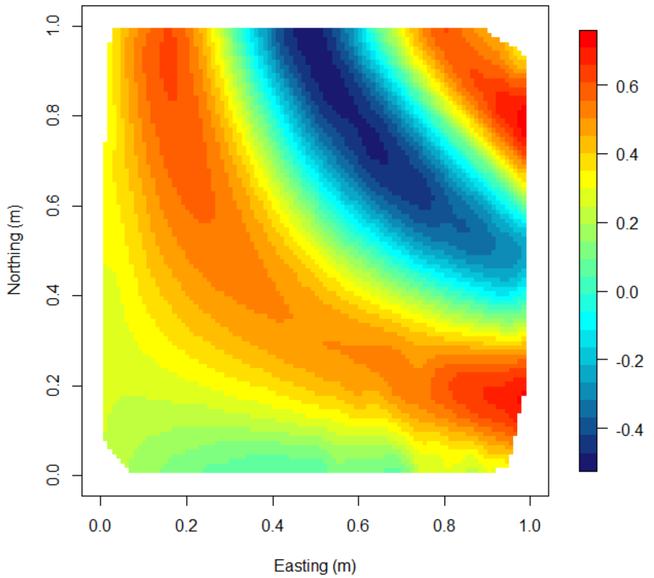
A flat semivariogram would suggest little spatial variation

variog command in the *geoR* R-package calculates empirical semivariograms

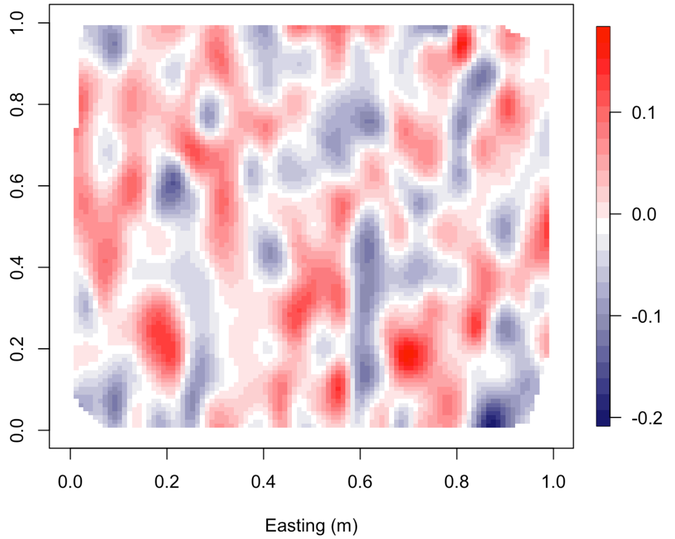
Dataset 1



$Y(s)$

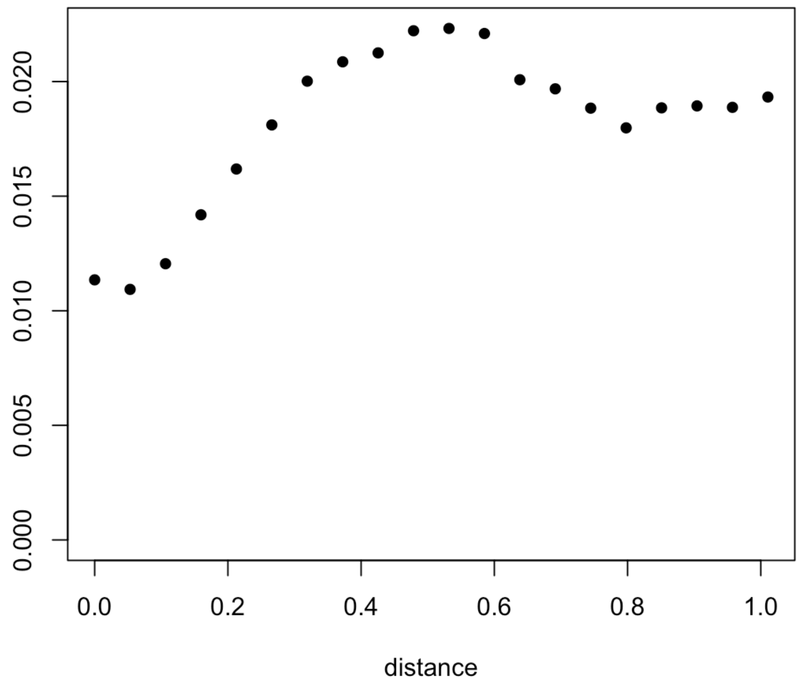


$X(s)$

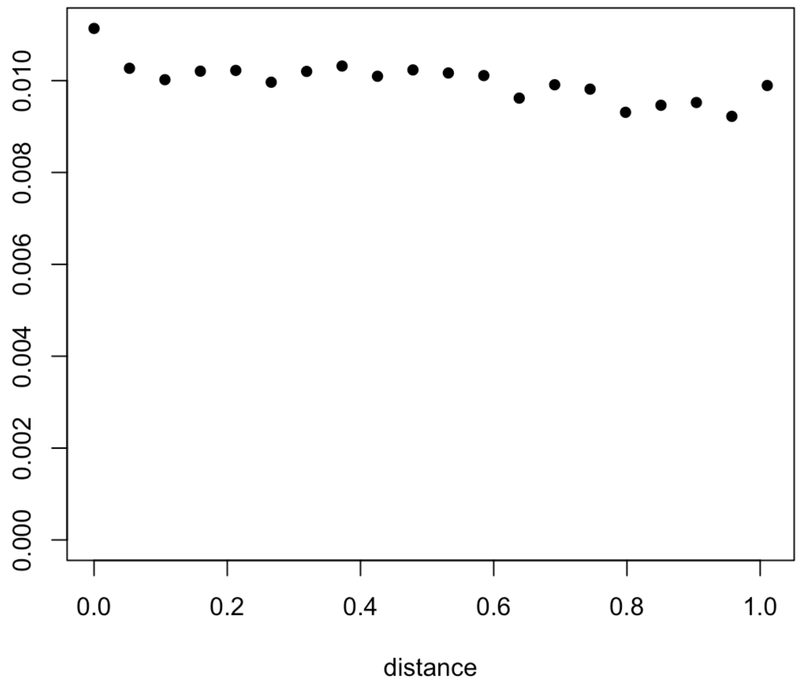


Residuals $Y(s) - \hat{Y}(s)$

Dataset 1



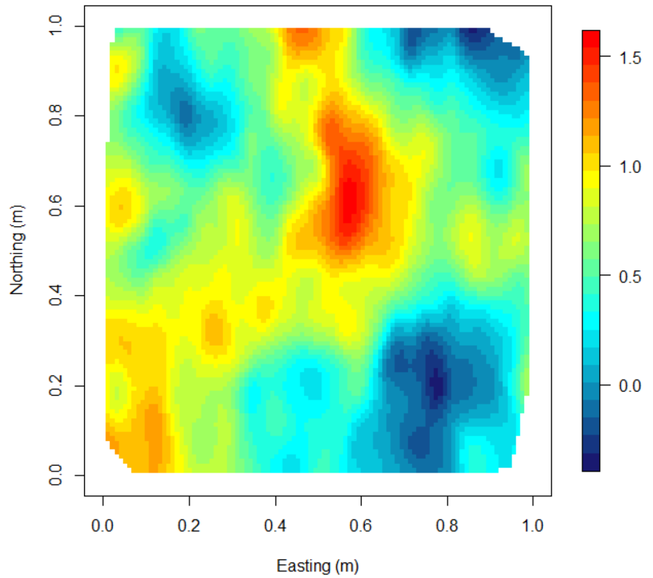
Variogram of $Y(s)$



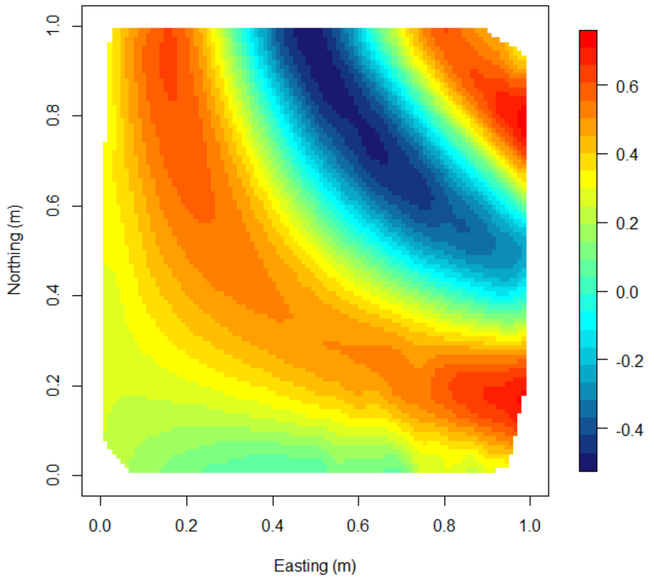
Variogram of residuals
$Y(s) - \hat{Y}(s)$

Variogram of residuals suggests very little spatial variation

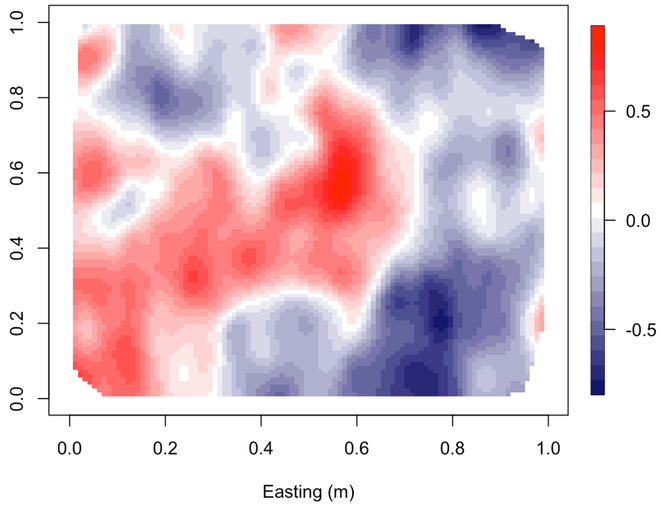
Dataset 2



$Y(s)$

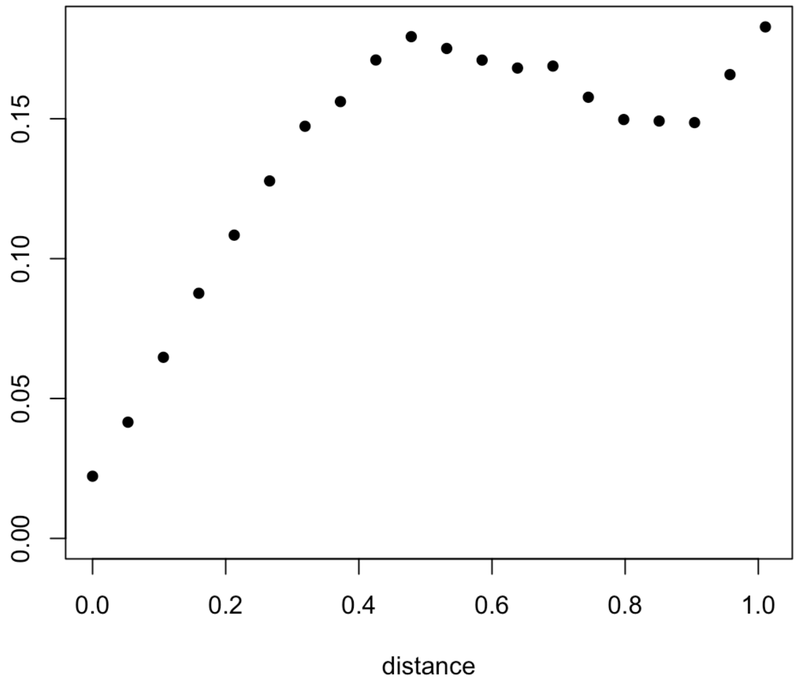


$X(s)$

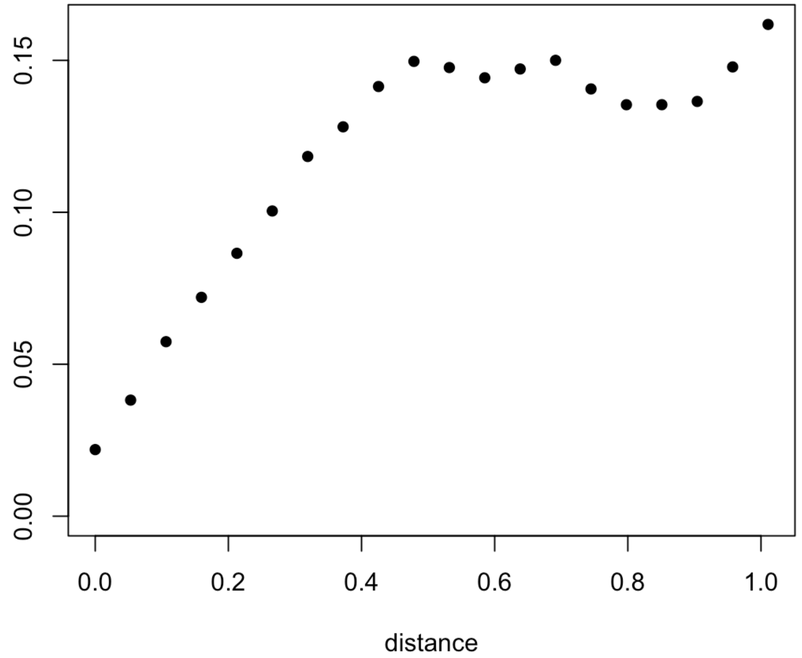


Residuals $Y(s) - \hat{Y}(s)$

Dataset 2



Variogram of $Y(s)$



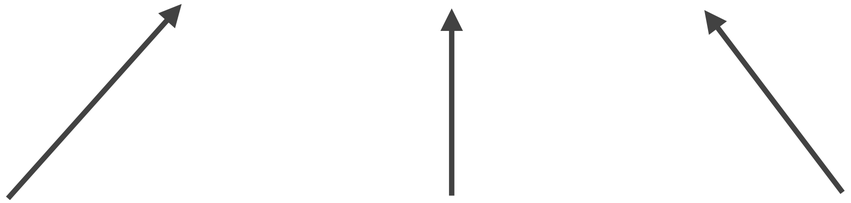
Variogram of residuals
$Y(s) - \hat{Y}(s)$

Variogram of residuals suggests residual spatial variation

Spatial linear mixed effect models (SLMM)

When purely covariate based models does not suffice, one needs to leverage the information from locations

SLMM: $Y(s_i) = X(s_i)'\beta + w(s_i) + \epsilon(s_i)$



Linear fixed effect

Spatial random effect

iid random errors

$w(s_i)$ is introduced to model spatial patterns in $Y(s_i)$ that are not explained by $X(s_i)$

Process-level model

Usually goal is predicting $Y(s)$ at any location s in the domain D

E.g., Conceptually pollutant level exists at all possible sites

SLMM: $Y(s) = X(s)'\beta + w(s) + \epsilon(s)$ for any location $s \in D$

Need to model $w(s)$ as a smooth function or stochastic process over D

Many approaches to model and estimate $w(s)$: basis function expansions, penalized regression splines, Gaussian Processes

Gaussian processes

$w(s)$ is often modeled as a Gaussian Process (GP)

$$w(\cdot) \sim GP(0, C(\cdot, \cdot))$$

$$w = w(S) = (w(s_1), \ldots, w(s_n))' \sim N(0, C)$$

$$C_{ij} = Cov(w(s_i), w(s_j)) = C(s_i, s_j)$$

Gaussian processes

$w(\cdot) \sim GP(0, C(\cdot, \cdot))$,

$w = w(S) = (w(s_1), \ldots, w(s_n))' \sim N(0, C)$

The covariance function models the spatial dependence

Stationarity and Isotropy: $C(s_i, s_j) = C(||h||)$ where $h = s_i - s_j$

Gaussian processes

$w(\cdot) \sim GP(0, C(\cdot,\cdot))$,

$w = w(S) = (w(s_1), \ldots, w(s_n))' \sim N(0, C)$

Matérn covariance function: A common, flexible family of covariances C specified using a spatial variance σ^2, spatial decay ϕ, and smoothness ν

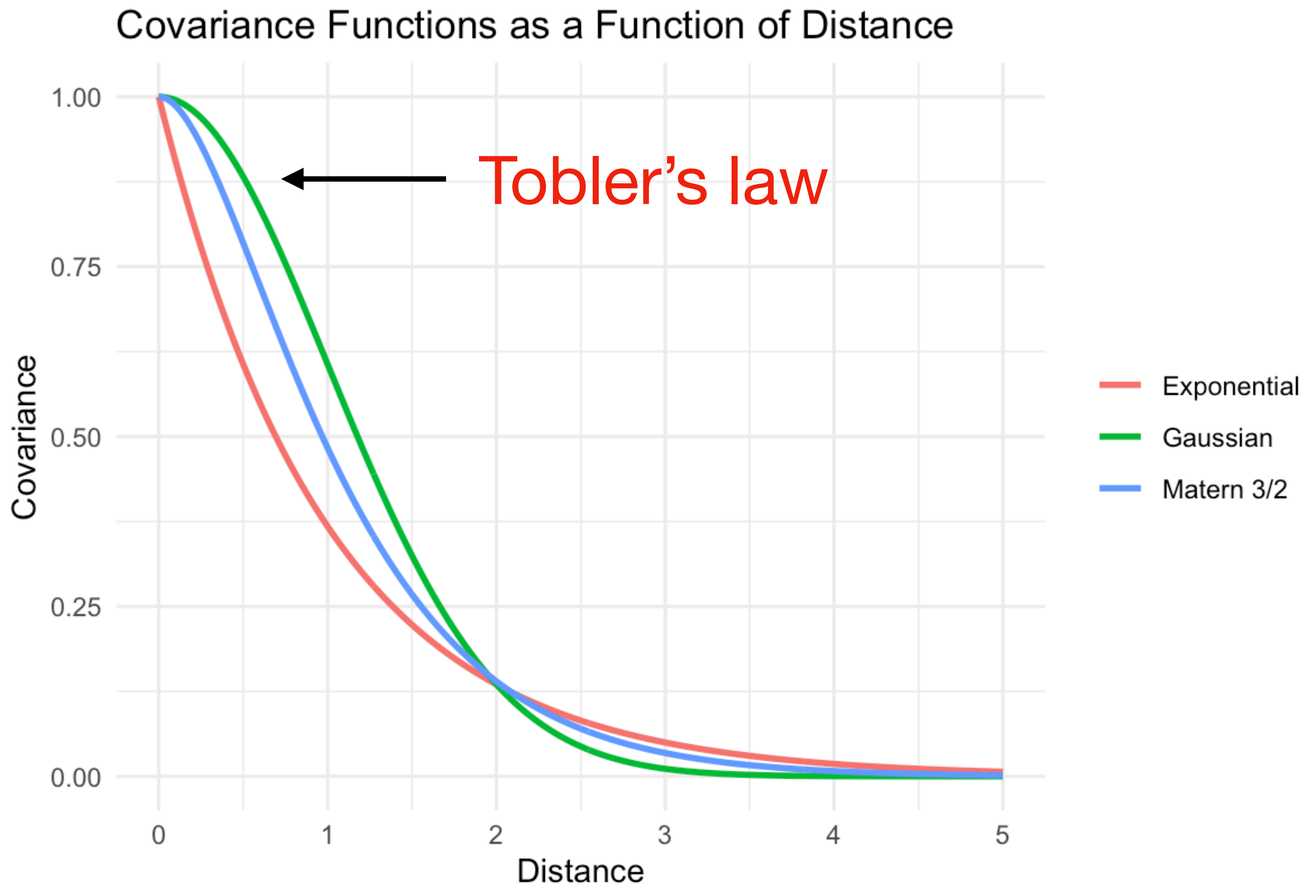
$\nu = 1/2$ (Exponential covariance): $C(||h||) = \sigma^2 \exp(-\phi||h||)$

$\nu = 3/2$: $C(||h||) = \sigma^2(1 + \phi||h||)\exp(-\phi||h||)$

$\nu = \infty$ (Gaussian covariance): $C(||h||) = \sigma^2 \exp(-\phi||h||^2)$

Gaussian processes

Matérn covariance function:



Gaussian processes

$$w(\cdot) \sim GP(0, C(\cdot, \cdot)),$$

$$w = w(S) = (w(s_1), \ldots, w(s_n))' \sim N(0, C)$$

Estimation of covariance parameters using Gaussian likelihood maximization

Process-level modeling: $w(s)$ is defined for any s in a region R

Allow predictions at any location s_0 via *kriging*

$$\begin{pmatrix} w(s_0) \\ w \end{pmatrix} \sim N\left(0, \begin{bmatrix} C(s_0, s_0) & C(s_0, S) \\ C(S, s_0) & C \end{bmatrix}\right)$$

$$w(s_0) \mid w \sim N(\mu(s_0), v(s_0))$$

Gaussian processes

$$w(\cdot) \sim GP(0, C(\cdot, \cdot)),$$

$$w = w(S) = (w(s_1), \ldots, w(s_n))' \sim N(0, C)$$

Flexibility and robustness: $w(s)$ for suitable covariance functions C can non-parametrically model any smooth fixed function $f(s)$ (van der Vaart 2008, 2011)

Spatial linear mixed effect models (SLMM)

SLMM: $Y(s_i) = X(s_i)'\beta + w(s_i) + \epsilon(s_i),\ i = 1,\ldots,n$

$w(\cdot) \sim GP(0, C(\cdot,\cdot))$, C is Matérn covariance with parameters σ^2, ϕ, ν

$w = w(S) = (w(s_1), \ldots, w(s_n))' \sim N(0, C)$

$\epsilon(s_i) \sim_{\text{iid}} N(0, \tau^2)$, τ^2 is often called the *nugget*

Spatial linear mixed effect models (SLMM)

Marginal model: $Y \sim N(X\beta, C + \tau^2 I)$

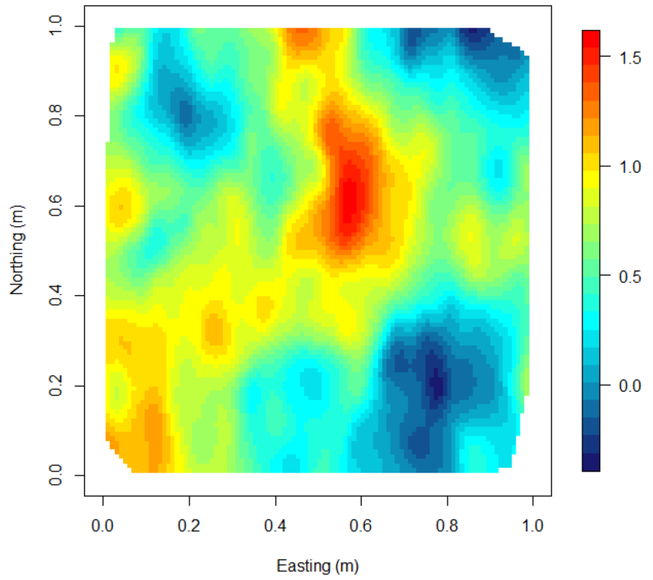
Parameters: Regression coefficient β and covariance parameters $\theta = (\tau^2, \sigma^2, \phi, \nu)$

Estimates using maximum likelihood estimation (MLE)

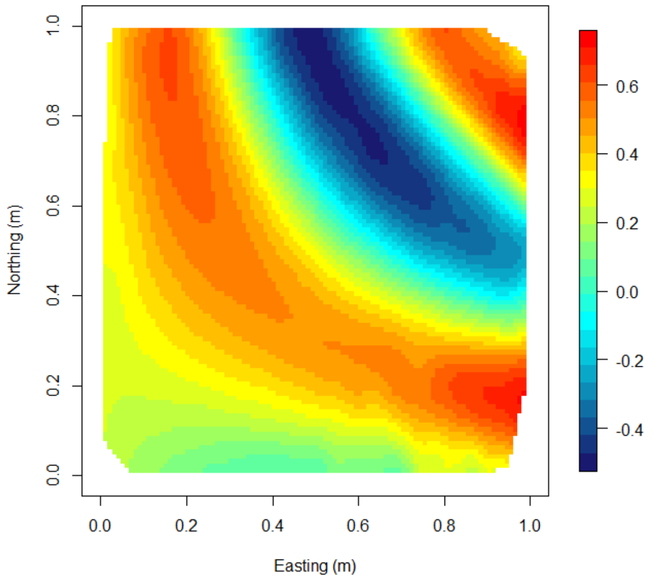
Predictions at a new location using kriging

Based on the conditional normal distribution of $Y(s_0) \mid Y$

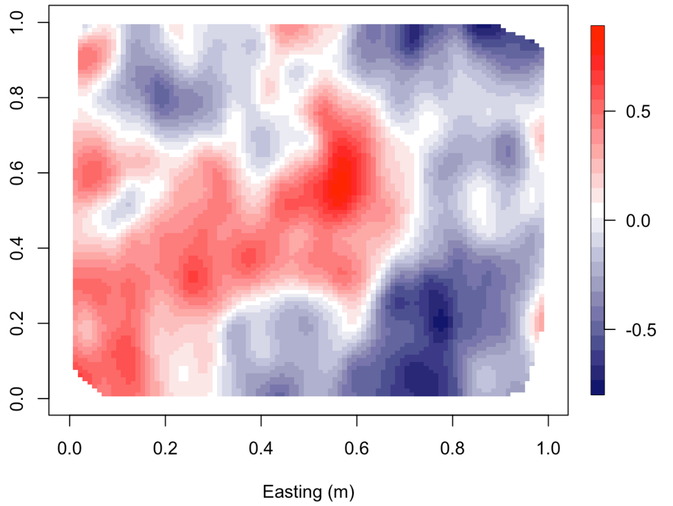
Dataset 2



$Y(s)$



$X(s)$



Residuals $Y(s) - \hat{Y}(s)$

Dataset 2

Model: $Y \sim N(X^*\beta^*, C + \tau^2 I), X^* = [1 : X], \beta^* = (\beta_0', \beta_1')'$

Parameters estimated using *likfit* function of geoR package

Parameter	Value
β_0	0.68
β_1	-0.50
τ^2	0.01
σ^2	0.14
ϕ [1]	0.26
ν [2]	0.5

[1] In geoR, ϕ is the inverse of our definition of ϕ [2] Fixed at 0.5, i.e., using the exponential covariance

Dataset 2

Model comparison metrics: Akaike Information Criterion (AIC) and Bayes Information Criterion (BIC)

- Lower values better
- AIC and BIC values are available from the output of likfit

	AIC	**BIC**
Model with Spatial	-146.1	-126.1
Model without Spatial	324.1	336.1

Spatial model is clearly favored

Dataset 2

Prediction: Available from krige.conv function of geoR

Data split:

80% for estimation of parameters (train),
20% for validation of predictions (test)

Dataset 2

Prediction metrics:

Root mean square prediction error (RMSPE) $= \sqrt{\frac{1}{n_{test}} \sum_{i=1}^{n_{test}} (y_i - \hat{y}_i)^2}$

Compares the point predictions $\hat{y}_i$

Lower values is better

Dataset 2

Prediction metrics:

Mean coverage probability (CP) of 95% prediction intervals

$$= \frac{1}{n_{test}} \sum_{i=1}^{n_{test}} I(y_i \in (\hat{y}_{i,0.025}, \hat{y}_{i,0.975}))$$

Evaluates the coverage of the interval predictions $(\hat{y}_{i,0.025}, \hat{y}_{i,0.975})$

Ideally should be close to 95%

Otherwise we will have under or over coverage

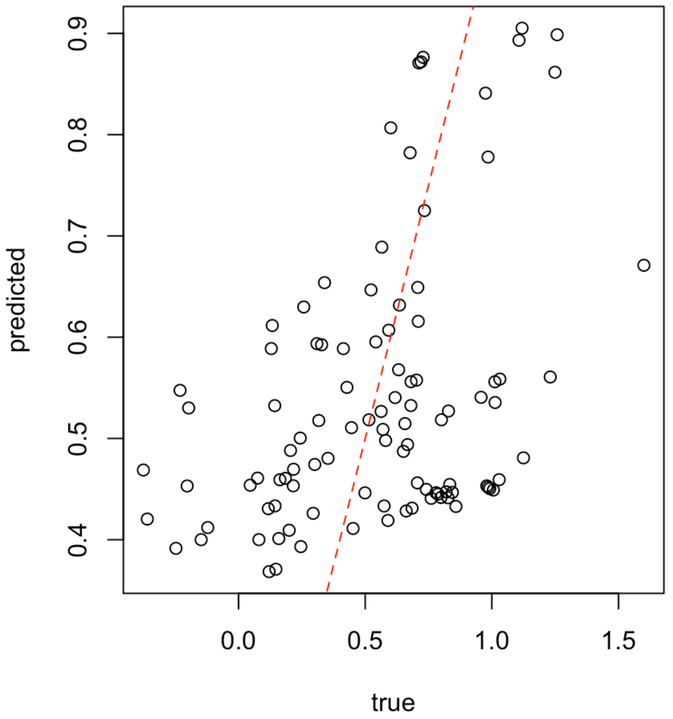
Mean prediction interval width (PIW) $= \frac{1}{n_{test}} \sum_{i=1}^{n_{test}} (\hat{y}_{i,0.975} - \hat{y}_{i,0.025})$

If CP ≈ 0.95, then smaller PIW is better

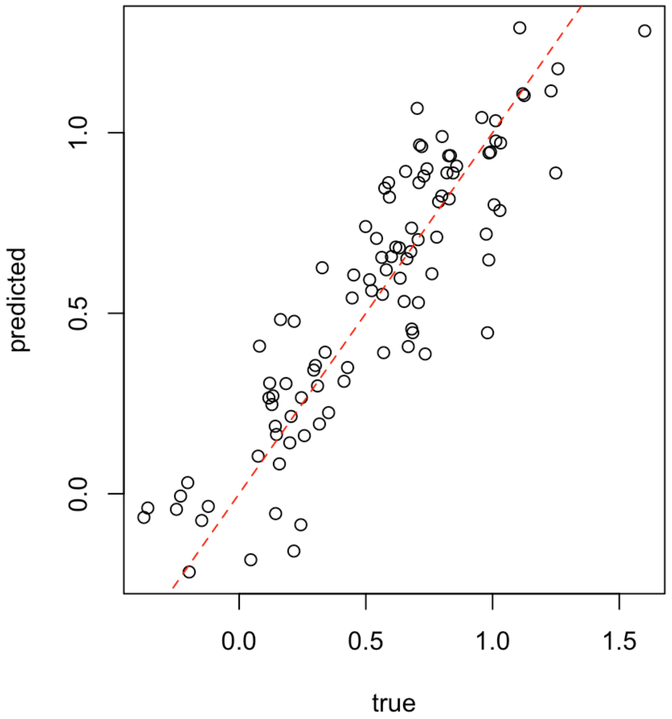
Dataset 2

Prediction:

Metric	Spatial	Non-Spatial
RMSPE	0.18	0.36
CP	0.95	0.95
PIW	0.68	1.42



Non-spatial model fit on test data



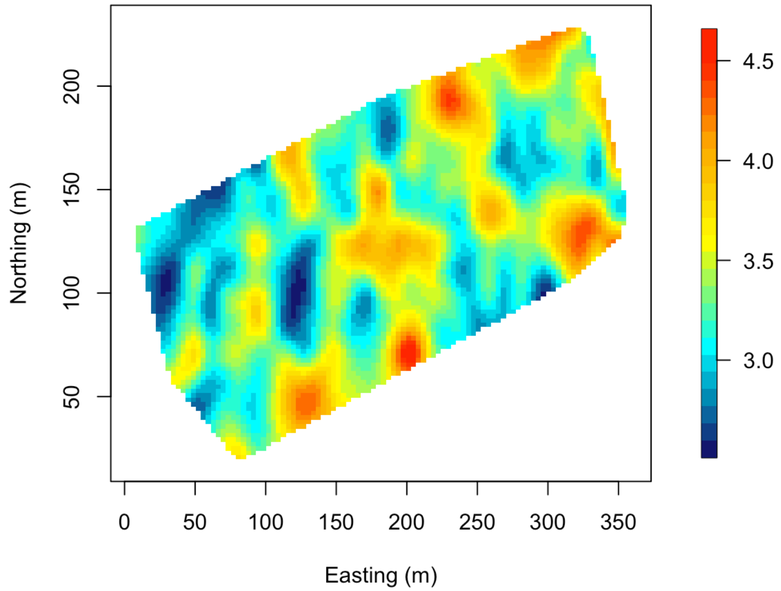
Spatial model fit on test data

Western Experimental Forestry (WEF) data

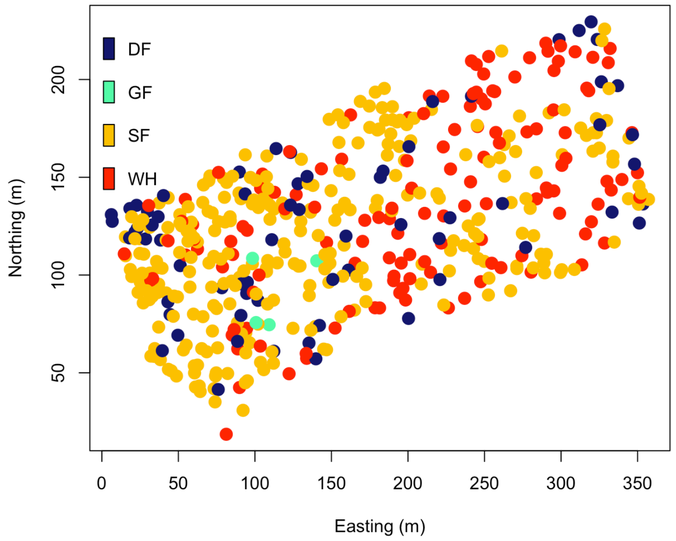
Data from *spBayes* package on census of all trees in a 10 ha. stand in Oregon

Response of interest: log(Diameter at breast height), i.e., log(DBH)

Covariate: Tree species (Categorical variable based on 4 species)



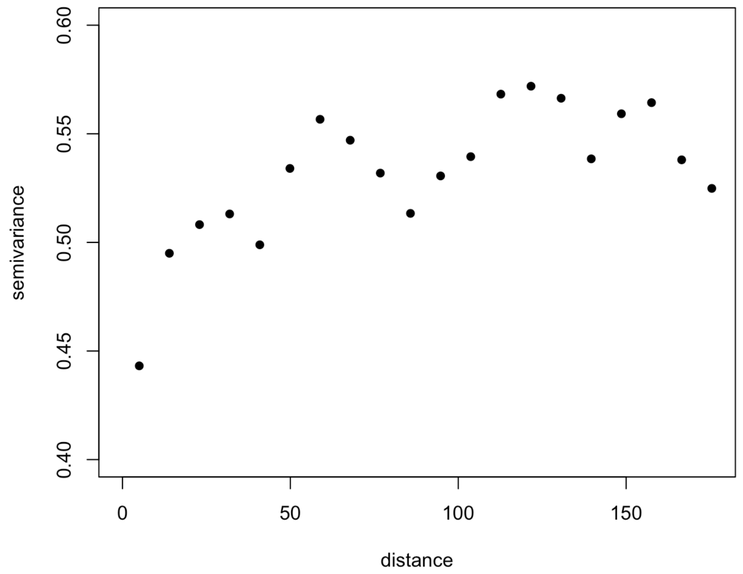
log(DBH)



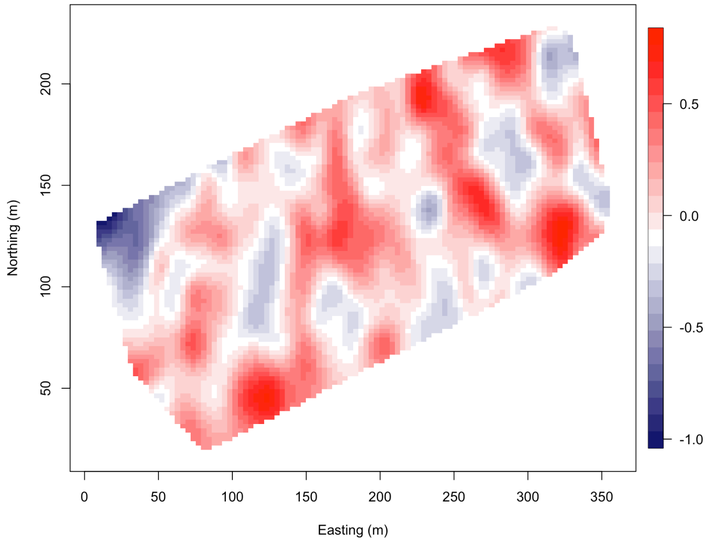
Species

Western Experimental Forestry (WEF) data

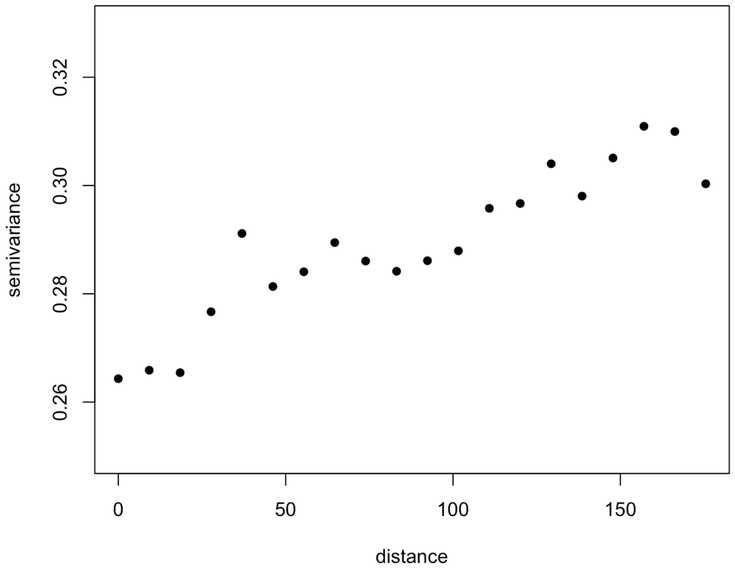
Linear model analysis:



Variogram of log(DBH)



Map of residuals

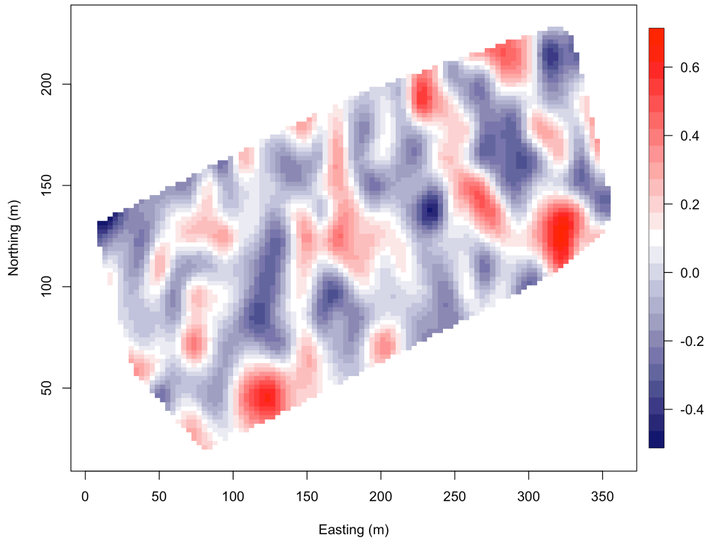


Variogram of residuals

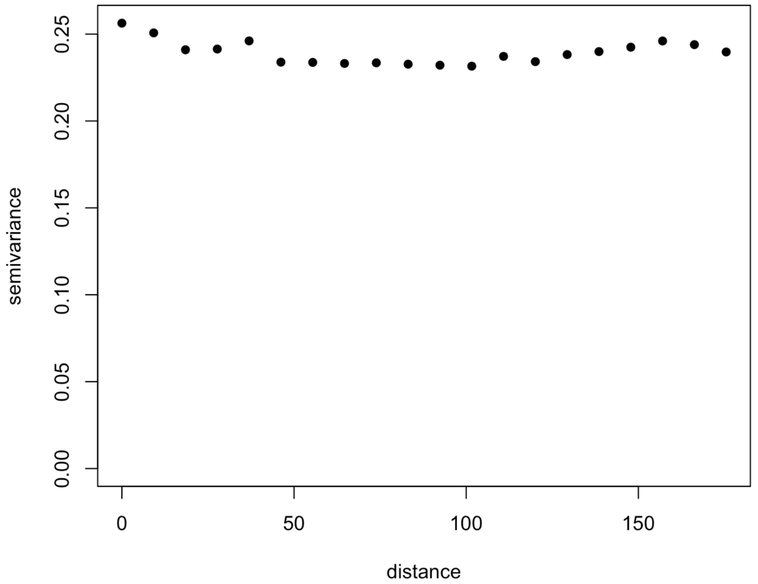
Western Experimental Forestry (WEF) data

Model comparisons:

Metric	Spatial	Non-Spatial
AIC	797	825
BIC	826	846



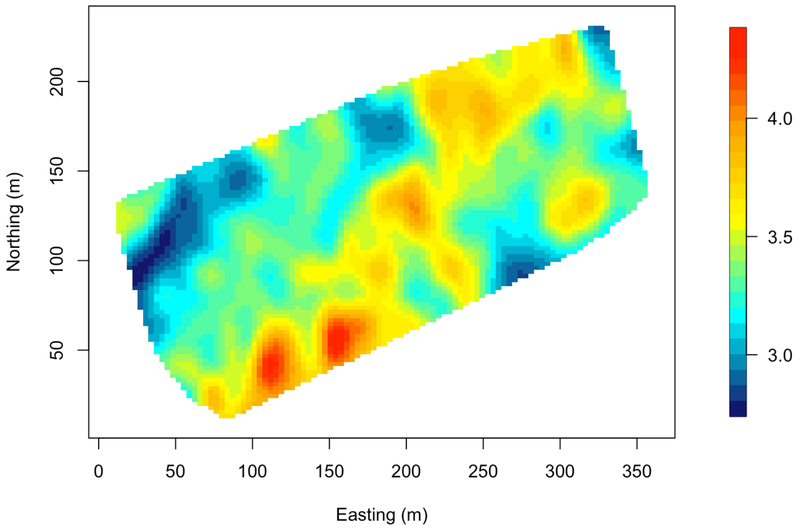
Map of residuals
from the spatial model



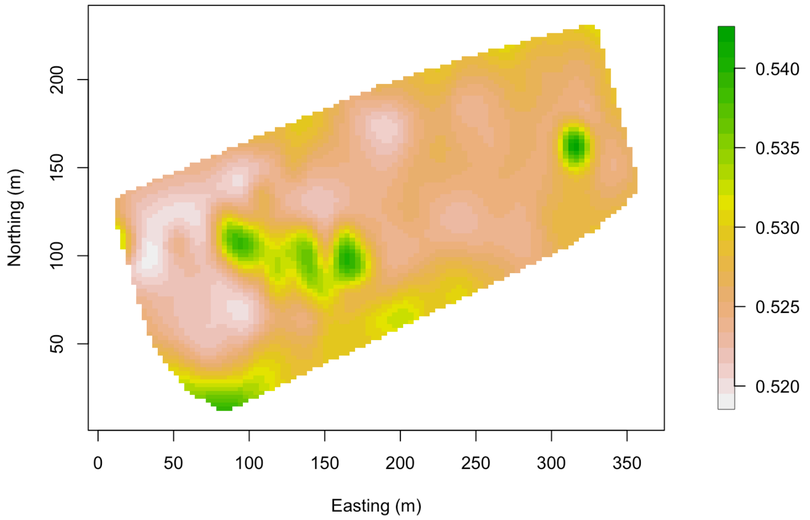
Variogram of residuals
from the spatial model

Western Experimental Forestry (WEF) data

Predictions:



Map of predicted DBH



Map of standard deviation of predicted DBH

Metric	Spatial	Non-spatial
RMSPE	0.54	0.56
CP	0.96	0.98
PIW	2.1	2.2

Big spatial data

Data at n locations: $S = \{s_1, \ldots, s_n\}$

Marginal model: $Y \sim N(X\beta, \Sigma(\theta))$ where $\Sigma(\theta) = C + \tau^2 I$

Parameter estimation using MLE

Log-likelihood: $l(\beta, \theta \mid Y) = -\frac{1}{2}\log\det(\Sigma(\theta)) - \frac{1}{2}(Y - X\beta)^\top \Sigma(\theta)^{-1}(Y - X\beta)$

Needs evaluation of $\det(\Sigma(\theta))$ and quadratic forms of $\Sigma(\theta)^{-1}$

Big spatial data

Prediction at a new location s_0: $Y(s_0) \mid Y, \theta, \beta = N\left(\mu(s_0), \sigma^2(s_0)\right)$

Conditional mean: $\mu(s_0) = X(s_0)^\top \beta + C(s_0, S)\Sigma^{-1}(Y - X\beta)$

Conditional variance: $\sigma^2(s_0) = C(s_0, s_0) + \tau^2 - C(s_0, S)\Sigma^{-1}C(S, s_0)$

Again needs evaluation of quadratic forms of Σ^{-1}

Computational details

$\Sigma := \Sigma(\theta)$ is a dense $n \times n$ matrix

Both $\det(\Sigma)$ and Σ^{-1} are best computed via the Cholesky decomposition

Cholesky decomposition: Any symmetric matrix A can be factorized as $A = LDL^\top$ where L is lower triangular and D is diagonal

Cholesky decomposition requires $O(n^2)$ storage and $O(n^3)$ time

Not feasible for large n

Methods for spatial big data

Low-rank models

Spectral approximations

Lattice-based methods

Multi-resolution approaches

Covariance tapering

Stochastic Partial Differential Equations

Nearest-neighbor models

See Heaton et al. (2019) for a review

Methods for spatial big data

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Nearest Neighbor Gaussian Processes

GP regression model: $Y \sim N(X\beta, \Sigma(\theta))$

Likelihood factorization: $p(Y) = p(Y_1) \times \prod_{i=2}^{n} p(Y_i \mid Y_1, \ldots, Y_{i-1})$

Vecchia's GP likelihood approximation (Vecchia, 1988, JRSSB):

$$p(Y) \approx p(Y_1) \times \prod_{i=2}^{n} p(Y_i \mid Y_{N(i)})$$

$N(i) =$ set of m nearest neighbors of location s_i among $s_1, \ldots, s_{i-1}$

Reduces computation time from $O(n^3)$ to $O(nm^3)$

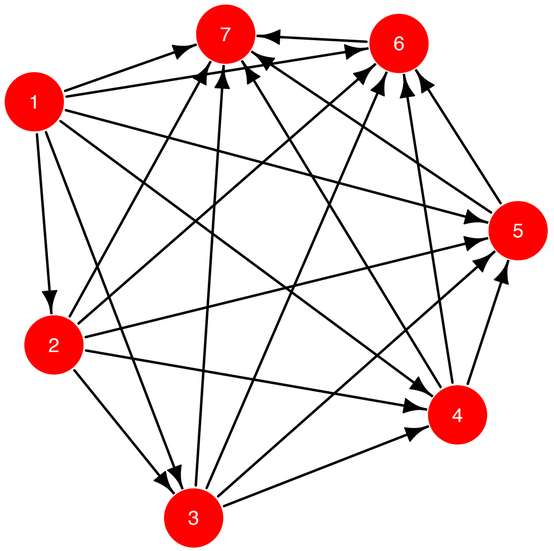
Nearest Neighbor Gaussian Processes

NNGP (Datta et al, 2016, JASA): Vecchia's approximation corresponds to a distribution $N(0,\tilde{\Sigma})$ and can be extended to a valid Gaussian process (NNGP)

Nearest Neighbor Gaussian Processes

NNGP (Datta et al, 2016, JASA): Vecchia's approximation is the likelihood of a distribution $N(0,\tilde{\Sigma})$ and can be extended to a valid Gaussian process (NNGP)

NNGP likelihood factorizes on a sparse directed acyclic graph (DAG)



Complete DAG

Full GP likelihood

$$p(y) = p(y_1)p(y_2 \mid y_1)$$

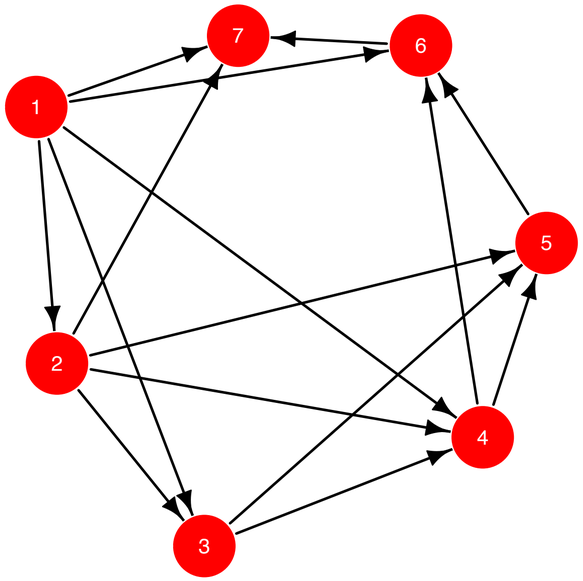
$$\times p(y_3 \mid y_1, y_2)p(y_4 \mid y_1, y_2, y_3)p(y_5 \mid y_1, y_2, y_3, y_4)$$

$$\times p(y_6 \mid y_1, y_2, y_3, y_4, y_5)p(y_7 \mid y_1, y_2, y_3, y_4, y_5, y_6)$$

Nearest Neighbor Gaussian Processes

NNGP (Datta et al, 2016, JASA): Vecchia's approximation is the likelihood of a distribution $N(0,\tilde{\Sigma})$ and can be extended to a valid Gaussian process (NNGP)

NNGP likelihood factorizes on a sparse directed acyclic graph (DAG)



3-NN DAG

NNGP likelihood

$$p(y) = p(y_1)p(y_2 \mid y_1)$$

$$\times p(y_3 \mid y_1, y_2)p(y_4 \mid y_1, y_2, y_3)p(y_5 \mid y_1, y_2, y_3, \cancel{y_4})$$

$$\times p(y_6 \mid \cancel{y_1}, \cancel{y_2}, y_3, y_4, y_5)p(y_7 \mid y_1, y_2, \cancel{y_3}, \cancel{y_4}, \cancel{y_5}, y_6)$$

Nearest Neighbor Gaussian Processes

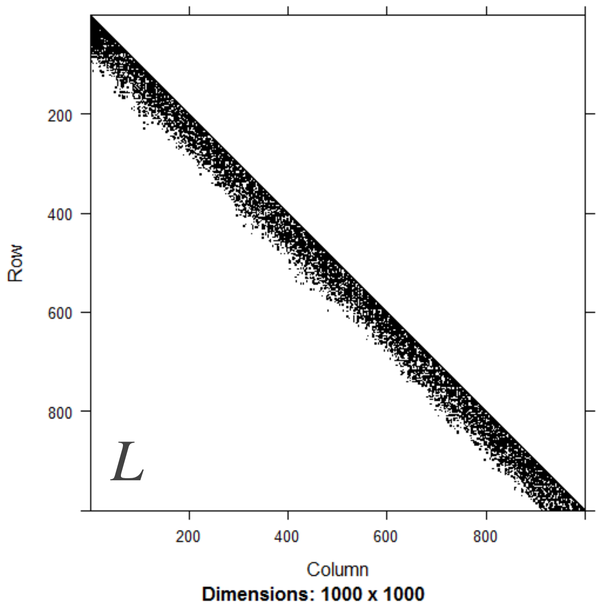
The NNGP precision matrix admits the factorization $\tilde{\Sigma}^{-1} = L^{\top} D L$

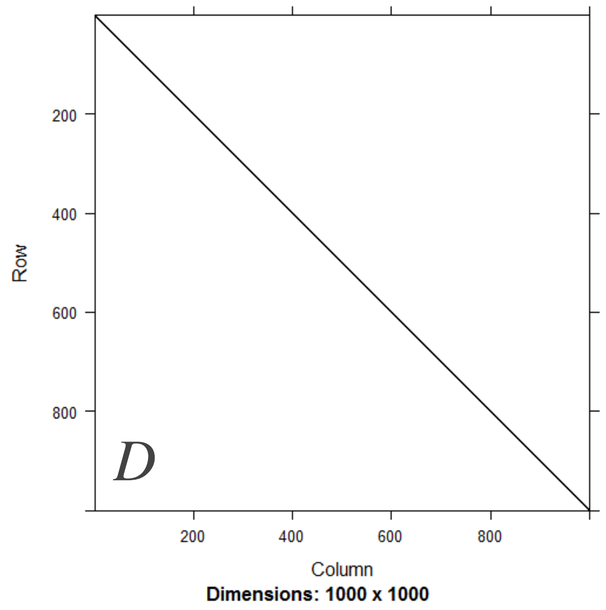
D is diagonal with entries d_i

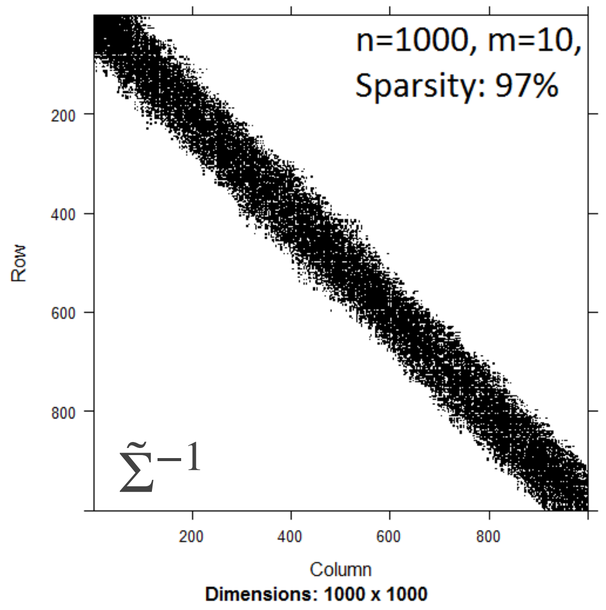
L is lower triangular and row sparse

Sparsity determined by the nearest-neighbor DAG

$\tilde{\Sigma}^{-1}$ is also sparse; total time to evaluate NNGP likelihood is $O(nm^3)$







Nearest Neighbor Gaussian Processes

Predictions:

NNGP prediction at a new location s_0:

$$Y(s_0) \mid Y = Y(s_0) \mid Y_{N_0}$$

$N_0 = m$ nearest neighbors of s_0 among $s_1, \ldots, s_n$

Nearest Neighbor Gaussian Processes

Software (R package):

BRISC (Saha and Datta)

- Frequentist (MLE) implementation
- Estimation with bootstrapped uncertainty
- Prediction with uncertainty
- Simulation of large spatial data

spNNGP (Finley, Datta, and Banerjee)

- Bayesian implementation
- Full posterior distributions using MCMC

Nearest Neighbor Gaussian Processes

Software (R package):

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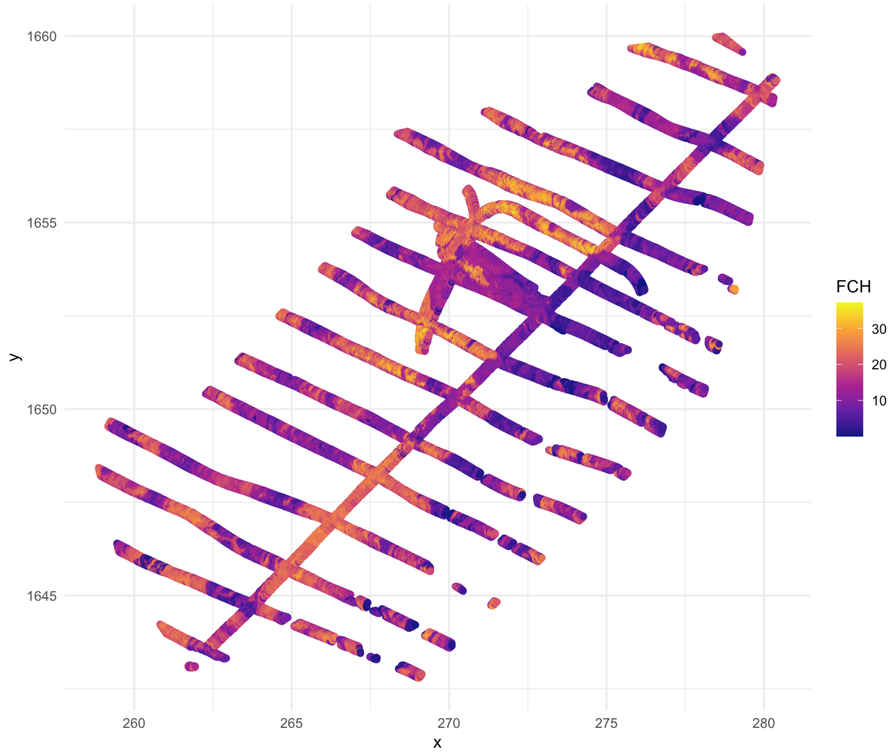
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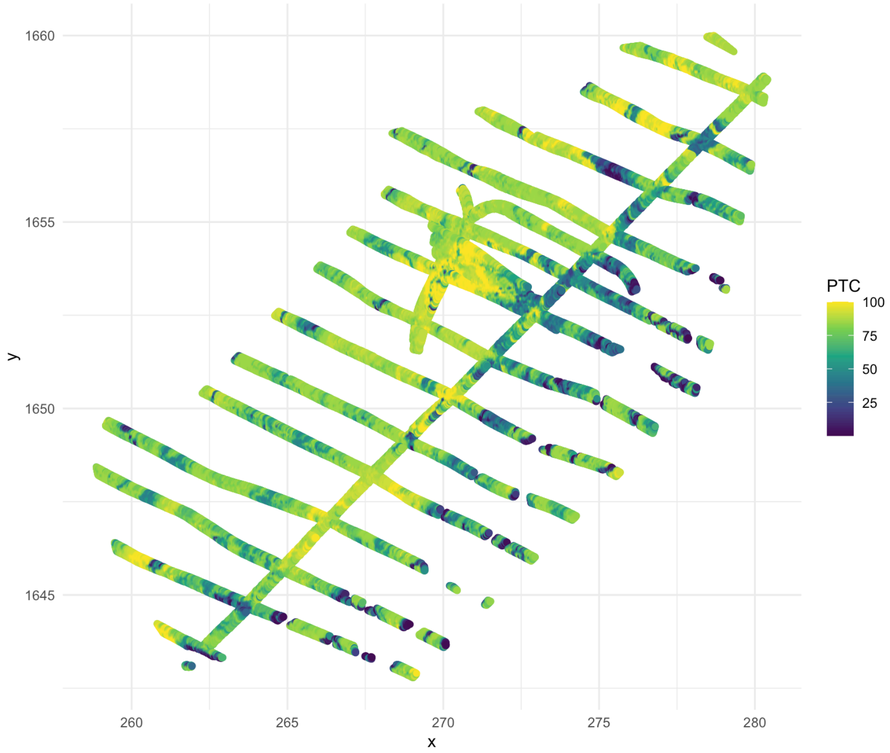
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Bonanza Creek Experimental Forest Data

Forest canopy height (FCH) estimates at 180,000 locations NASA Goddard's LiDAR in Bonanza Creek Experimental Forest, Alaska



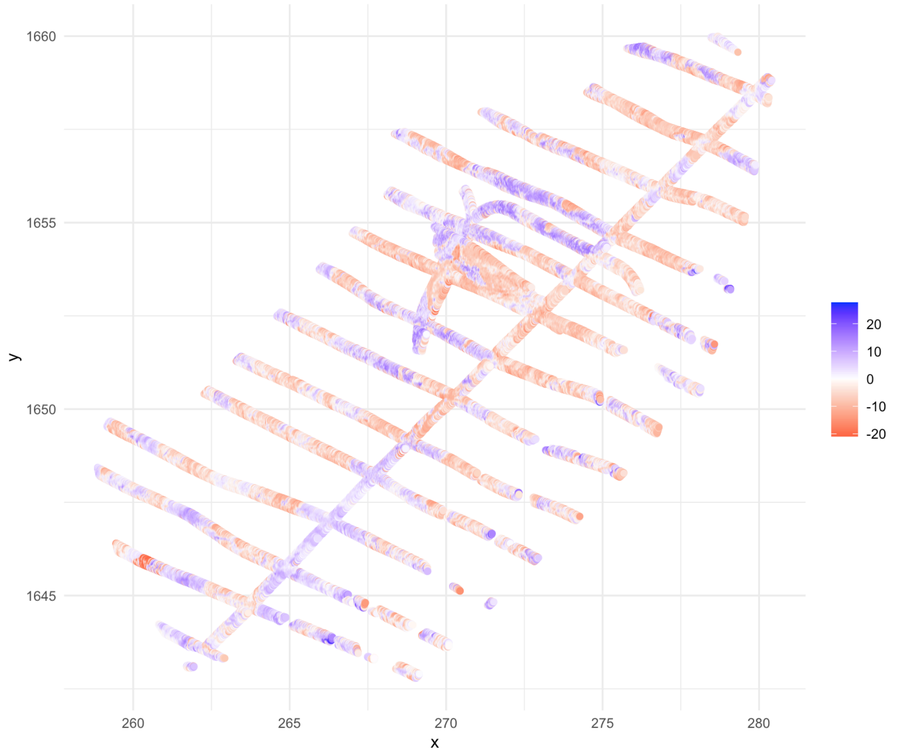
Forest canopy height (FCH)



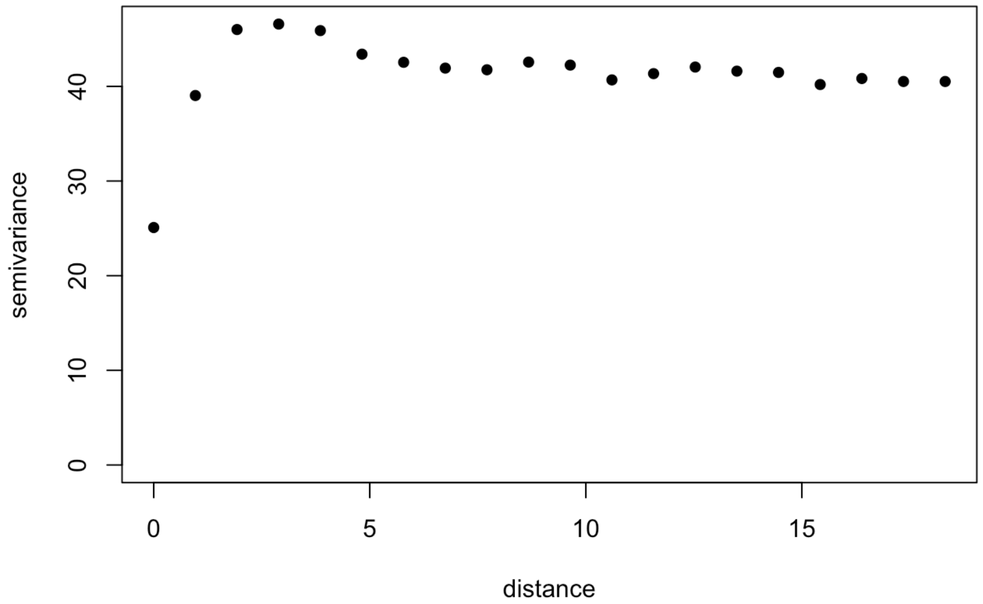
Covariate — Percent tree cover (PTC)

Bonanza Creek Experimental Forest Data

Linear model:



Residuals



Variogram of a subset of residuals

Bonanza Creek Experimental Forest Data

Comparison of *BRISC_estimation and geoR:likfit* on a subset of size 5000

Quantity	BRISC	geoR
Intercept	8.64	8.70
PTC	0.09	0.09
σ^2	37.82	37.83
τ^2	8.00	8.09
ϕ	2.92	2.87
Runtime	0.8 secs	116.2 secs

Bonanza Creek Experimental Forest Data

Spatial model:

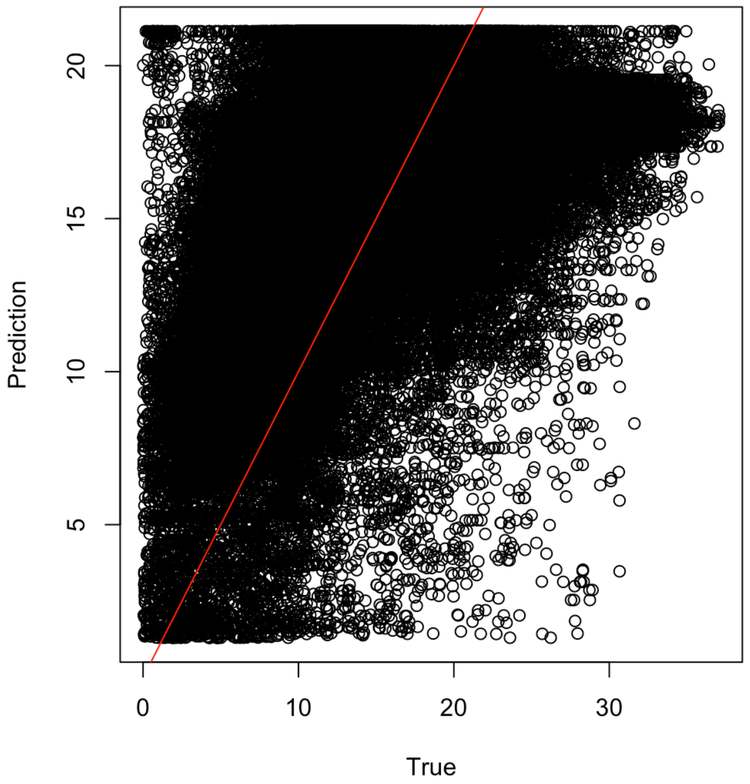
Training data of size 100000

Model fitting using *BRISC_estimation*

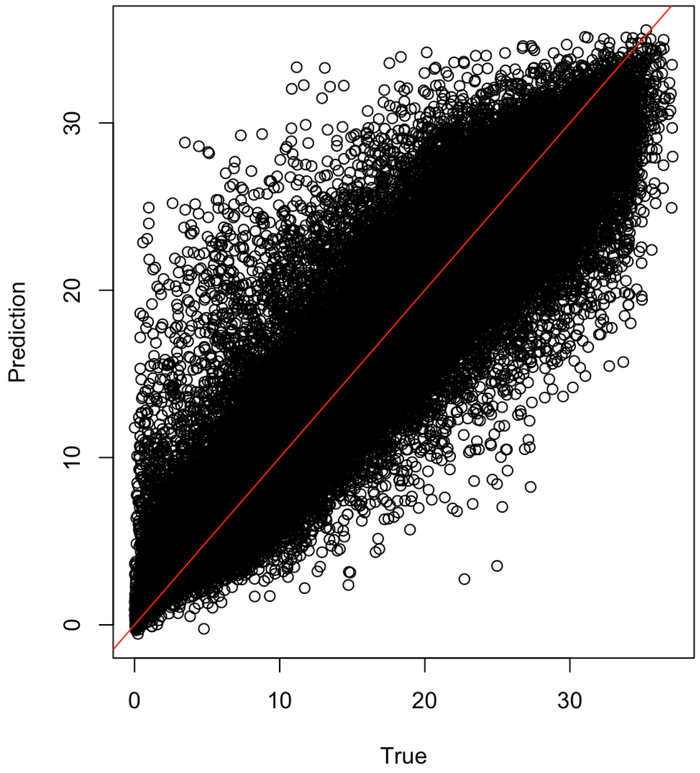
Predictions using *BRISC_prediction*

Run time: 9.94 secs

Metric	Spatial	Non-Spatial
RMSPE	2.92	6.59
CP	0.94	0.96
CIW	11.15	25.84



Non-spatial model



Spatial model

Bonanza Creek Experimental Forest Data

Spatial model:

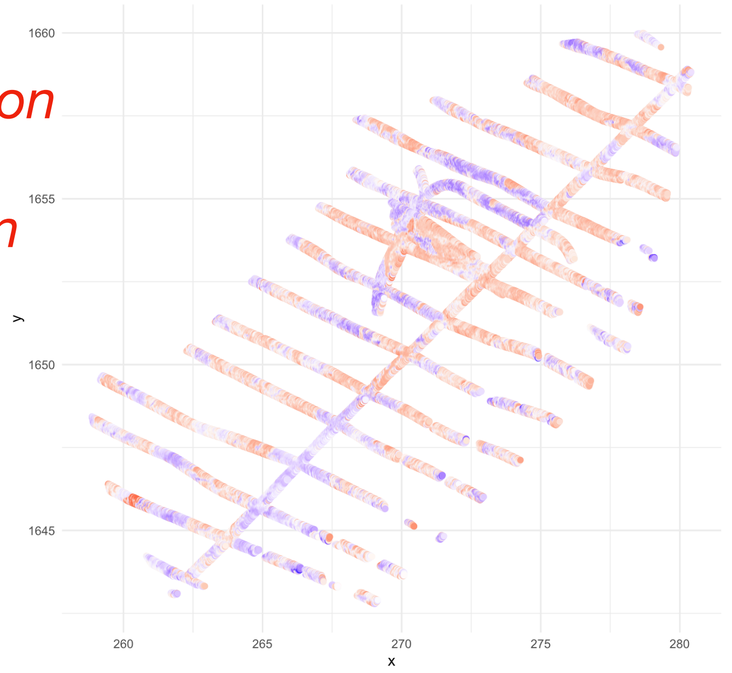
Training data of size 100000

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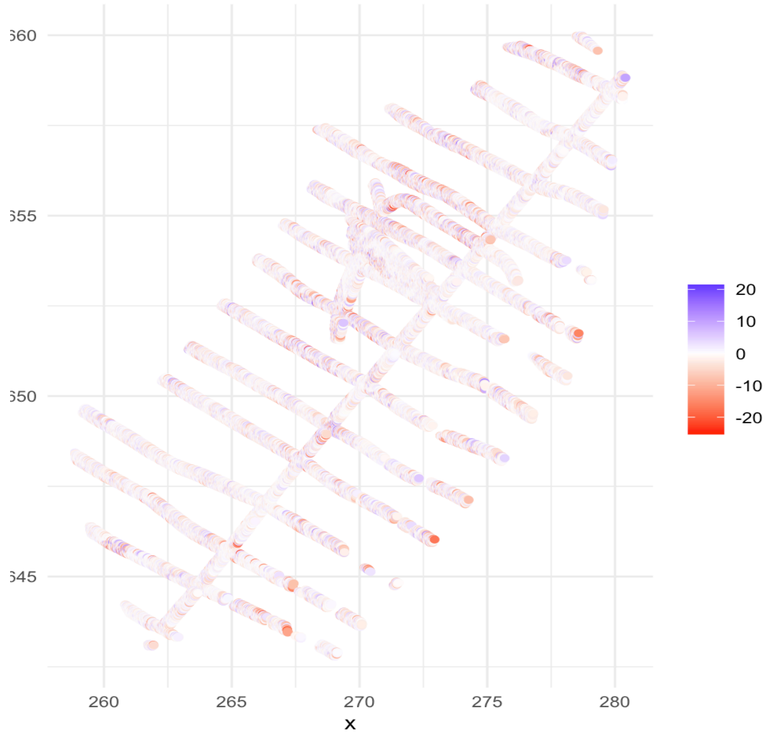
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Spatial
model residuals



Non-spatial
model residuals

Summary

Introduction to geostatistics
- Data setup and analysis objectives

Exploratory data analysis to understand need for spatial modeling
- Maps and variograms of data and linear model residuals

Spatial linear mixed effect models
- Process level modeling and Gaussian processes
- Parameter estimation
- Prediction (kriging) with uncertainty quantification

Summary

Model comparison
- Estimation: AIC, BIC
- Prediction: RMSPE, coverage probability and width of prediction intervals,

Big spatial data
- Computing challenges
- Fast alternatives (Nearest Neighbor Gaussian Process)

Spatial analysis using geoR and BRISC R-packages

References

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